



TIPS & TRICKS

How to check the type of file system on an unmounted device

Here are a few simple steps for checking the file system type of an unmounted device.

Run the following command in the terminal:

```
# fdisk /dev/sdX -l
```

This will display details that give you a basic idea of the file system's structure. The output is something like what's shown below:

```
Device Boot Start End Blocks Id System
/dev/sdb1 1 986 7912515+ b W95FAT32
```

Next, you can try the following command:

```
# blkid
```

The output is something like what follows:

```
/dev/sdb1: LABEL="HP V220w" UUID="74C8-177C" TYPE="vfat"
```

You can also use *parted* as shown below:

```
# parted /dev/sdX -l
```

This is similar to *fdisk* but the output is more formatted as shown below:

```
Model: hp v220w (scsi)
Disk /dev.sdb : 8103MB
```

```
Number Start End Size Type FileSystem
1 979KB 8103MB 8102MB primary fat32
```

—Sharmin Shaikh, imsharmeen@gmail.com

Find out your hard disk's serial number without physical access to it

Here is a trick to find out your hard disk's serial number in Linux when physical access to the disk is not possible. You need to run the following command in the terminal to get these details:

```
#hdparm -I /dev/sda
```

To get only the serial number, you can run the following command:

```
#hdparm -I /dev/sda | grep Serial
```

—Suresh Jagtap, smjagtap@gmail.com

Access control lists to secure the file/directory

The access control list is used to define the permissions for the file and directory on an individual level. It allows you to give permissions for any user or group to any disk resource.

Thus, it provides more security. Rules must be specified in the following formats.

1. To add the permissions to the user, use the following code:

```
setfacl -m "u:username:permissions"
or
setfacl -m "u:username:permissions"
```

2. To add permissions to the group, use:

```
setfacl -m "g:groupname:permissions"
or
setfacl -m "g:gid:permissions"
```

3. To remove all permissions, use:

```
setfacl -b
```